

PHYS 4027 / 5027 • FALL 2026 • LECTURE 5

Numerical Integration



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Today



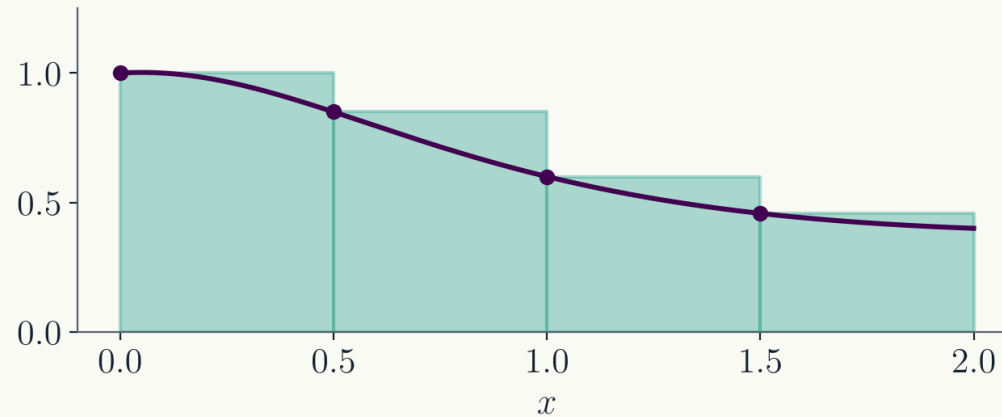
- **Rules on a grid.** The trapezoidal rule and Simpson's rule.
- **Adaptive refinement.** Choosing the number of points automatically.
- **Gauss quadrature.** Choosing the points themselves.

Part I · Rules on a grid



Trapezoidal · Simpson

Starting from the definition



The Riemann sum with N intervals of width $h = (b - a)/N$:

$$\int_a^b f(x) dx \approx \sum_{i=0}^{N-1} f(x_i) h$$

This converges, but the error shrinks only as h .

Riemann sum error



- On one interval, the rectangle ignores the variation of f . By the mean value theorem, $f(x) - f(x_i) = f'(\xi)(x - x_i)$ for some ξ between x_i and x , so one interval contributes error

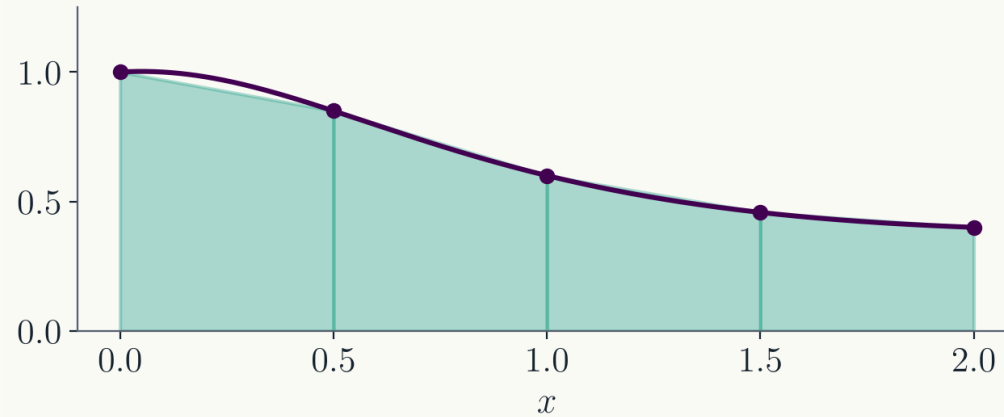
$$\int_{x_i}^{x_{i+1}} [f(x) - f(x_i)] dx = \frac{1}{2} f'(\xi_i) h^2.$$

- There are $N = (b - a)/h$ intervals. Summing N terms of size h^2 costs one power of h :

$$\varepsilon = \frac{h}{2} \sum_i f'(\xi_i) h \approx \frac{h}{2} \int_a^b f' dx = \frac{h}{2} [f(b) - f(a)].$$

- **Notation.** $\varepsilon = O(h^p)$ means $|\varepsilon| \leq Ch^p$ as $h \rightarrow 0$, with C independent of h . Halving h divides the error by 2^p ; on a log-log plot against N , the slope is $-p$.

Trapezoidal rule



Replace f on each interval by its *linear interpolation*, and integrate:

$$\int_a^b f(x) dx \approx \left[\frac{1}{2} f(x_0) + f(x_1) + \cdots + f(x_{N-1}) + \frac{1}{2} f(x_N) \right] h$$

Interior points are shared by two trapezoids and get full weight; the endpoints get half.

Trapezoidal rule: error on one interval

One interval, centered at m , width h . Expand f around m :

$$f(x) \approx f(m) + f'(m)(x - m) + \frac{1}{2} f''(m)(x - m)^2.$$

integral:
$$\int_{m-h/2}^{m+h/2} f(x) dx \approx f(m) h + \frac{f''(m)}{24} h^3$$

trapezoid:
$$\frac{h}{2} \left[f\left(m - \frac{h}{2}\right) + f\left(m + \frac{h}{2}\right) \right] \approx f(m) h + \frac{f''(m)}{8} h^3$$

(plug $x = m \pm h/2$ into the expansion; the f' terms cancel in the sum)

Subtract: one interval is off by $\left(\frac{1}{24} - \frac{1}{8}\right) f''(m) h^3 = -\frac{1}{12} f''(m) h^3.$

Trapezoidal rule: total error

Add up the $N = (b - a)/h$ intervals, with midpoints $m_1, \dots, m_N$:

$$\varepsilon \approx \sum_{i=1}^N \left(-\frac{f''(m_i)}{12} h^3 \right) = -\frac{h^2}{12} \sum_{i=1}^N f''(m_i) h \approx -\frac{h^2}{12} \int_a^b f''(x) dx$$

- The last sum is a Riemann sum, so

$$\varepsilon \approx \frac{h^2}{12} [f'(a) - f'(b)].$$

- N terms of size h^3 give $O(h^2)$: the same bookkeeping as the rectangle rule, one order better.

The error comes entirely from the curvature: straight lines ($f'' = 0$) are integrated exactly.

The Euler–Maclaurin formula

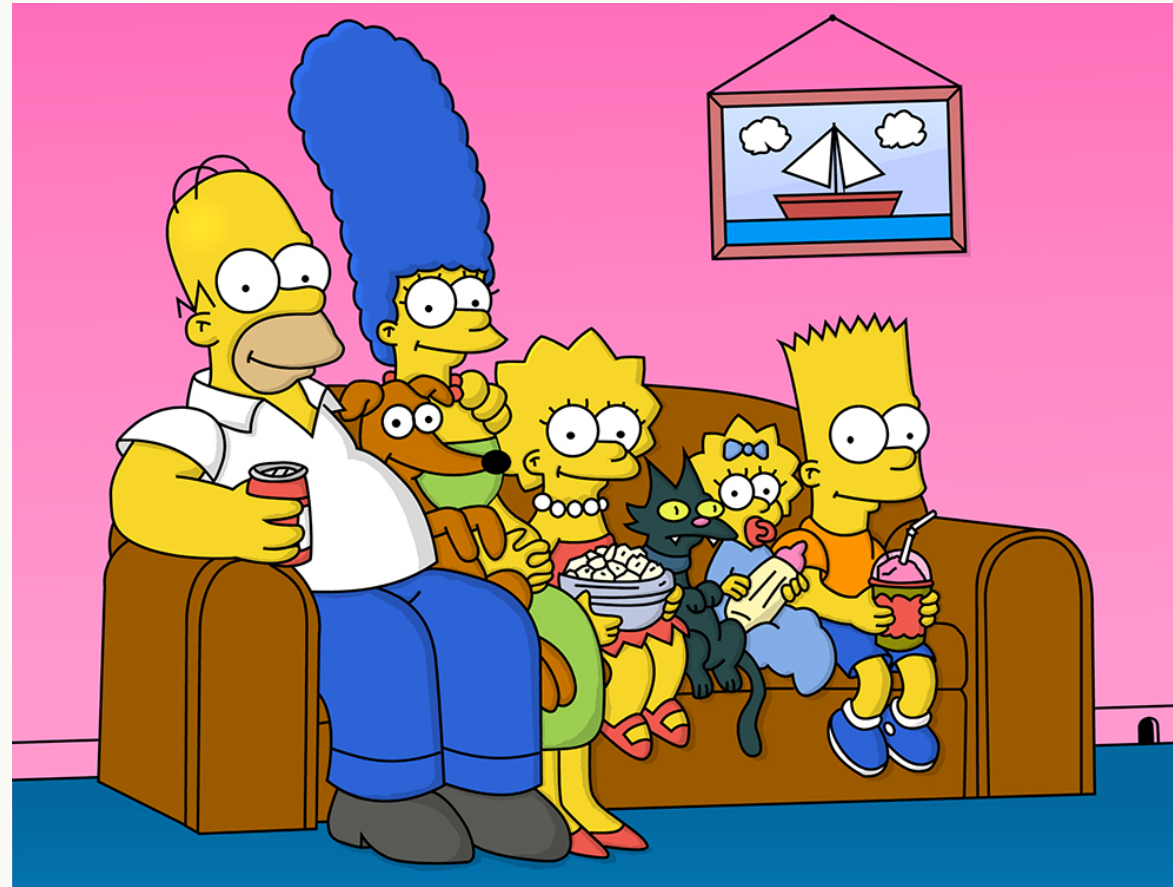


The trapezoidal error has an exact expansion, the *Euler–Maclaurin formula*:

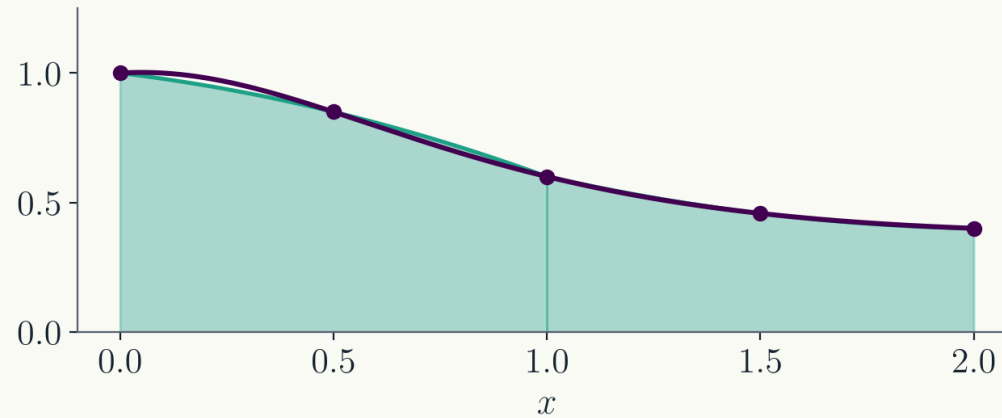
$$\varepsilon = \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)!} h^{2k} \left[f^{(2k-1)}(a) - f^{(2k-1)}(b) \right]$$

- B_{2k} are the Bernoulli numbers ($B_2 = \frac{1}{6}$ gives the $\frac{1}{12}$).
- Only *even* powers of h appear. Removing the h^2 term therefore improves the error to h^4 , two orders at once.

Simpson's rule



Simpson's rule



Interpolate a *parabola* through each pair of intervals and integrate it. With N even:

$$\int_a^b f(x) dx \approx \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 4f(x_{N-1}) + f(x_N)]$$

The pattern 1, 4, 2, 4, ..., 2, 4, 1: each parabola spans two intervals and weights its midpoint by 4.

Simpson's rule: the error



- The error is

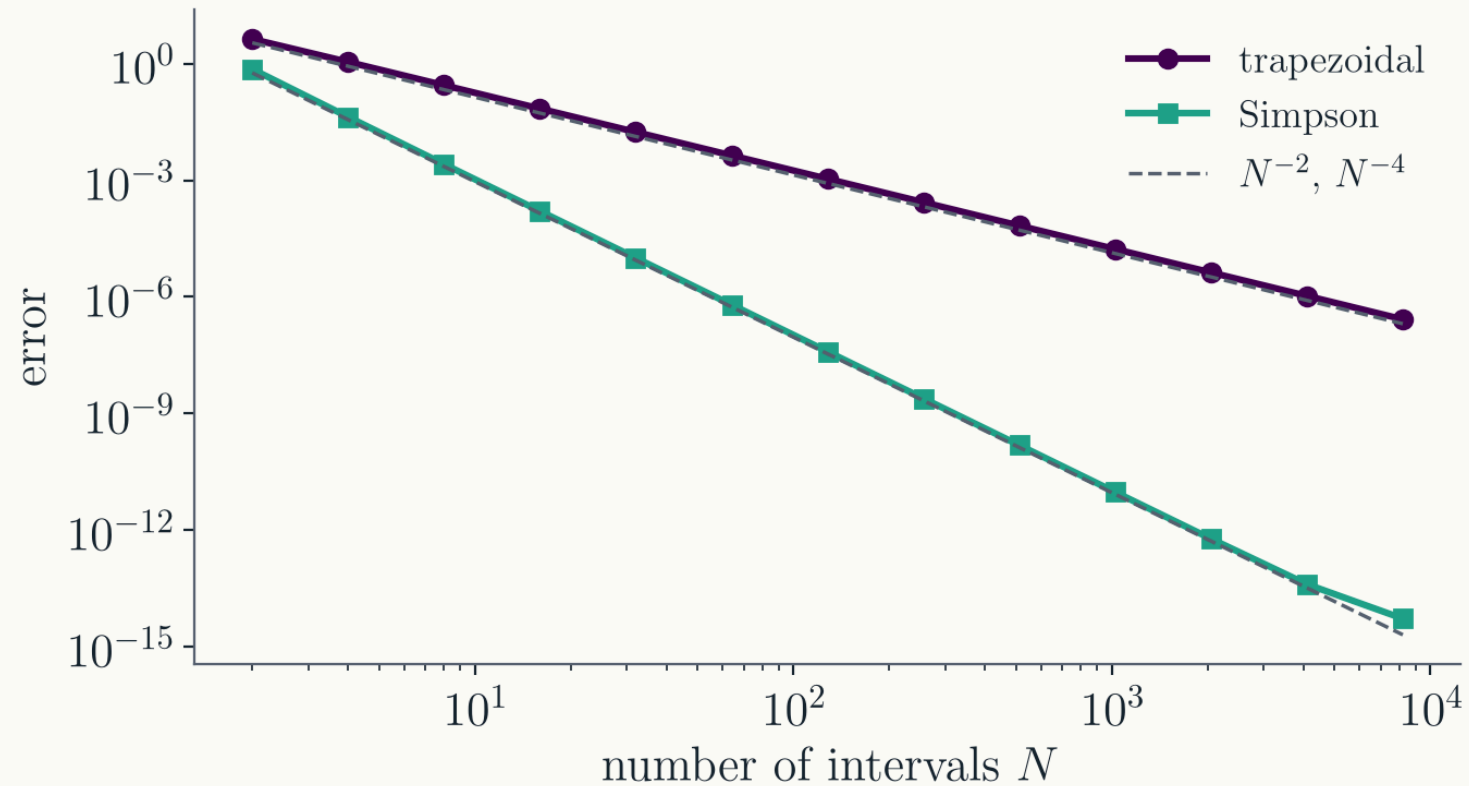
$$\varepsilon = \frac{h^4}{180} [f'''(a) - f'''(b)] + O(h^6),$$

two orders better than trapezoidal, for the same number of function evaluations.

- A parabola guarantees exactness for quadratics. Simpson's rule is in fact exact for *cubics* as well.
- Why? A cubic minus its interpolating parabola is odd about the midpoint of the pair; its integral vanishes by symmetry.

Measured convergence

Both rules on $\int_0^2 x^4 \sinh^{-1}(x) dx = 8.15336 \dots$, against the exact value:



- Measured slopes: -1.99 and -4.02 , matching the h^2 and h^4 error formulas.

Part II · Adaptive refinement



How many points are enough?

Refine until it stops changing



The right N is usually not known in advance. Instead, keep doubling it, reusing the values already computed:

$$\begin{aligned} S_1 &= \frac{1}{2} [f(a) + f(b)] h_1, & h_1 &= b - a \\ S_2 &= \frac{1}{2} S_1 + f(a + h_2) h_2, & h_2 &= \frac{1}{2} h_1 \\ S_3 &= \frac{1}{2} S_2 + [f(a + h_3) + f(a + 3h_3)] h_3, & h_3 &= \frac{1}{2} h_2 \end{aligned}$$

Each level evaluates f only at the new midpoints; no value is computed twice.

Estimating the error



- The trapezoidal error is $\varepsilon \approx \frac{h^2}{12} [f'(a) - f'(b)]$, and the boundary term does not change under refinement: halving h divides the error by 4.
- So the difference between consecutive levels estimates the error:

$$\varepsilon_n \approx \frac{1}{3} (S_n - S_{n-1})$$

- Refine until $|\varepsilon_n|$ (or $|\varepsilon_n/S_n|$) drops below your tolerance.

Cancel the error instead



If S_{n-1} carries error ε and S_n carries $\varepsilon/4$, a linear combination can cancel ε entirely:

$$S'_n = \frac{4S_n - S_{n-1}}{3}$$

What does this do?

Writing S'_n out in terms of the function values shows that it is *exactly Simpson's rule* on the finer grid.

- Its error estimate: $\varepsilon_n \approx \frac{1}{15} (S'_n - S'_{n-1})$.
- Strategy: run the doubling trapezoidal levels $S_1, S_2, \dots$, form S'_n at each level, stop when the estimate is below tolerance.

Implementation

```
template <typename F>
std::tuple<bool, double> adaptive_simpson(const F& f, double a, double
                                         double tol) {
    // repeat the doubling trapezoidal levels, form S' at each level,
    // stop when the error estimate drops below tol
}
```

- Same conventions as the root finders: `{converged, value}`, a maximum refinement depth, `false` if the tolerance is never reached.
- The tolerance discussion from root finding applies unchanged: absolute or relative, and there is a floor set by rounding.

Part III · Gauss quadrature



Choosing the nodes

Moving the nodes



- Every rule so far has the shape

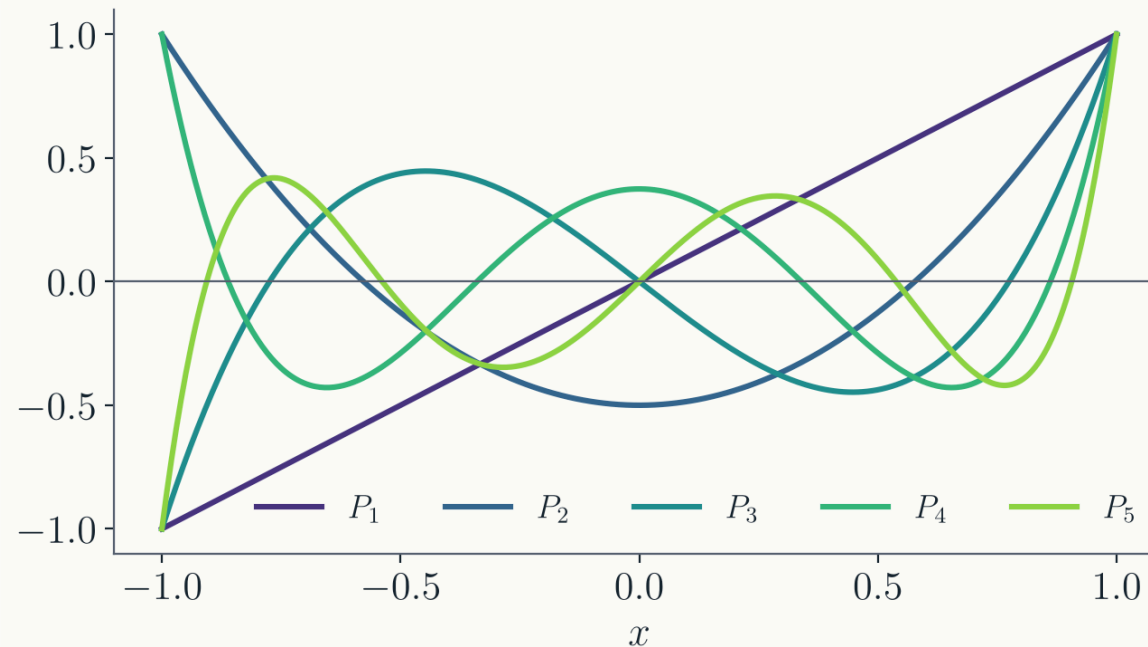
$$\int_a^b f(x) dx \approx \sum_{i=0}^{n-1} w_i f(x_i),$$

with the x_i fixed on an even grid. Rules of that family are called *Newton–Cotes* formulas.

- Now let the x_i move. That doubles the free parameters: n weights *and* n node positions, $2n$ in total.
- A polynomial of degree $2n - 1$ has $2n$ coefficients. Chosen well, n evaluations can integrate every such polynomial *exactly*.

With $n = 5$, five function values can integrate any degree-9 polynomial exactly.

Gauss–Legendre quadrature



- On $[-1, 1]$: the nodes x_i are the n roots of the Legendre polynomial $P_n(x)$, and the weights are

$$w_i = \frac{2}{(1 - x_i^2)[P'_n(x_i)]^2}.$$

Nodes and weights, $n = 10$



$\pm x_i$	w_i
0.1488743389816312	0.2955242247147529
0.4333953941292472	0.2692667193099963
0.6794095682990244	0.2190863625159821
0.8650633666889845	0.1494513491505806
0.9739065285171717	0.0666713443086881

- The nodes come in $\pm$ pairs with equal weights, and none of them are endpoints. They cluster toward the endpoints, where interpolation error is largest.
- In practice: tabulated, or computed once by Newton's method on $P_n(x) = 0$.

Any interval



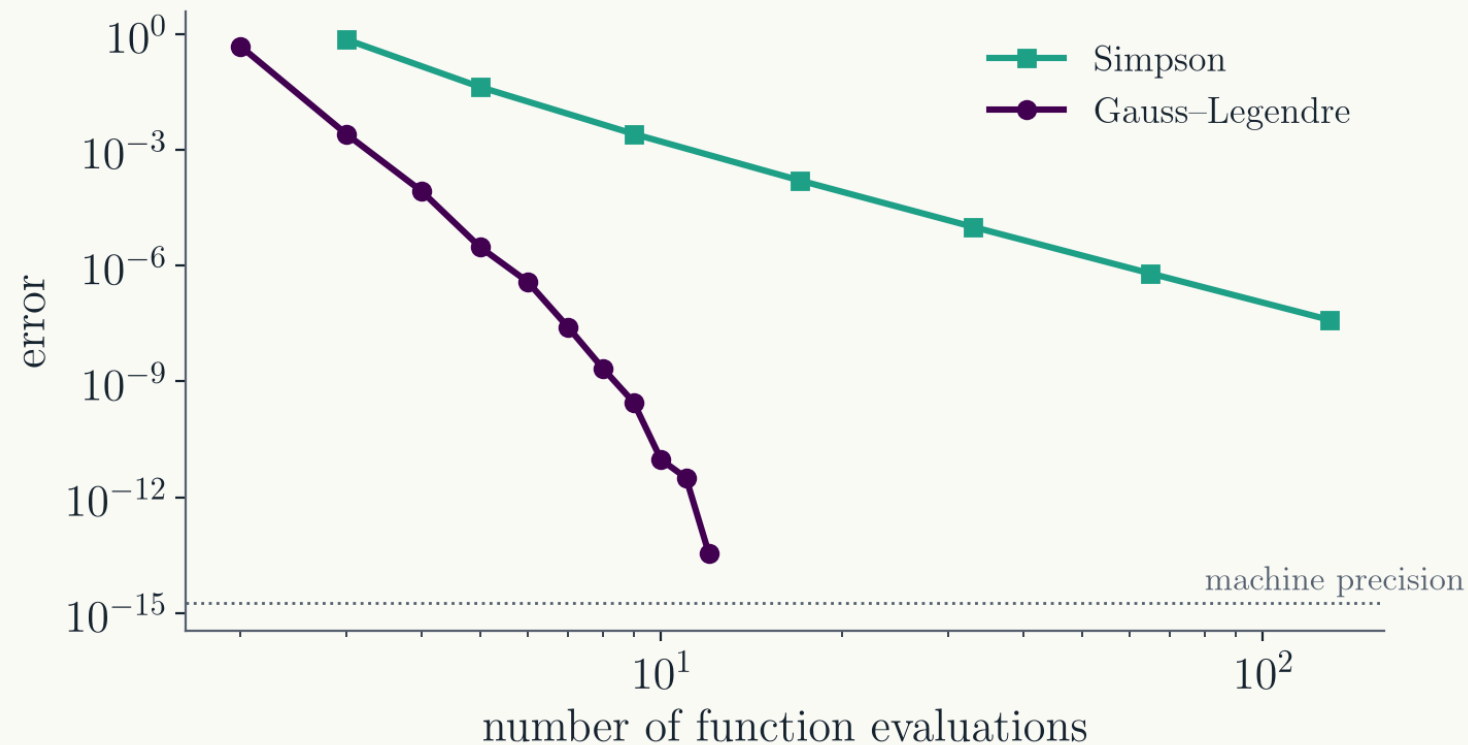
Gauss–Legendre lives on $[-1, 1]$. For $[a, b]$, substitute $x = \frac{1}{2}(b - a)t + \frac{1}{2}(b + a)$:

$$\int_a^b f(x) dx = \frac{b - a}{2} \int_{-1}^1 f(x(t)) dt \approx \frac{b - a}{2} \sum_{i=0}^{n-1} w_i f(x(t_i))$$

A linear change of variable and one overall factor.

Convergence of Gauss quadrature

The same test integral, $\int_0^2 x^4 \sinh^{-1}(x) dx$, cost measured in function evaluations:



- With 12 evaluations the error is 4×10^{-14} ; Simpson would need about 10^5 intervals. The error falls geometrically with n , not as a power of h .

Why it is exact at degree $2n - 1$



Take any polynomial f of degree $\leq 2n - 1$ and divide by P_n :

$$f(x) = P_n(x) q(x) + r(x), \quad \deg q, \deg r \leq n - 1$$

- **The q piece integrates to zero:** P_n is orthogonal to every polynomial of lower degree, so $\int_{-1}^1 P_n(x) q(x) dx = 0$.
- **The rule never sees q :** at the nodes, $P_n(x_i) = 0$, so $f(x_i) = r(x_i)$.
- **The r piece is what the sum computes:** $\deg r \leq n - 1$, so r is completely determined by its n values $r(x_i)$ (Lecture 4). Its exact integral is therefore a weighted sum of those values, $\int_{-1}^1 r(x) dx = \sum_i w_i r(x_i)$, and that is what defines the weights w_i .

Drawbacks of Gauss quadrature

- **Non-smooth integrands.** The derivation assumed f is well approximated by a high-degree polynomial. A kink, a singularity ($1/\sqrt{x}$), or a discontinuity breaks the geometric convergence.
- **Oscillatory integrands.** $\sin(1/x)$ near 0 is not well approximated by any polynomial.
- **No cheap error estimate.** Different n share no nodes, so estimating the error requires a full recomputation.

Use Gauss quadrature for smooth integrands; use adaptive Simpson when f is poorly understood or an error estimate is needed.

Where this goes next



- **Next lecture:** Solving ODEs, Runge-Kutta methods.
- References:
 - Numerical Recipes, 3rd ed., Chapter 4 (quadrature).
 - Burden & Faires, *Numerical Analysis*, Chapter 4.